'Assignment-01'

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Sec: 10

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Question-1

(a)

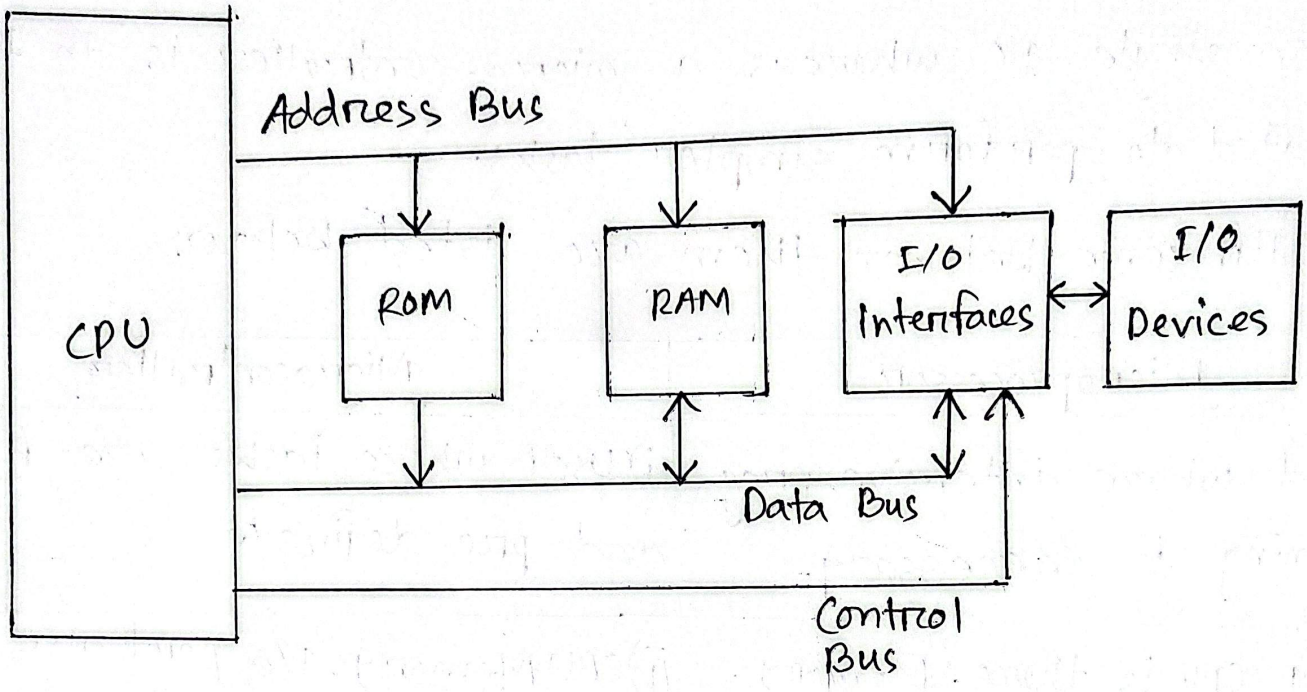
A microprocessor incorporates most or all of the functions of a CPU on single IC whereas a micro-controller is an IC dedicated to perform simpler tasks.

The difference between them are listed below:

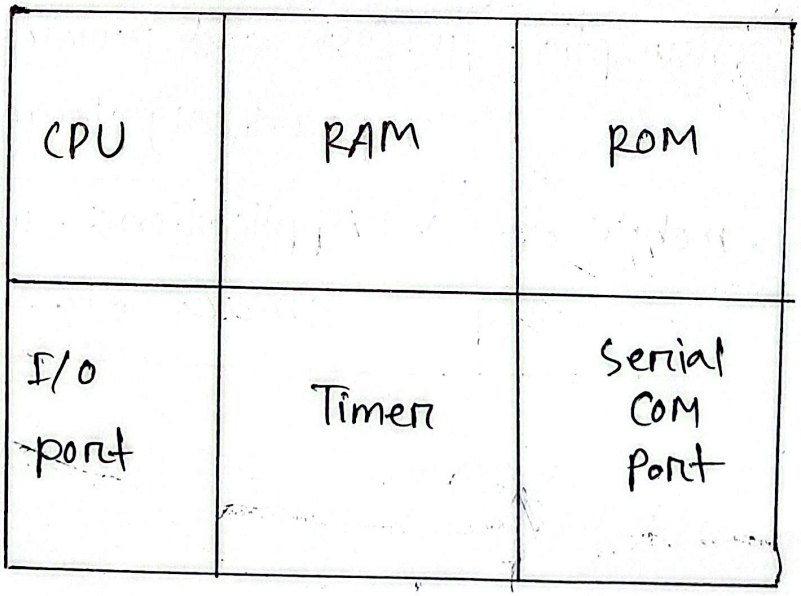
Microprocessor	Microcontroller
i) Used where intensive programming is necessary.	i) Used where tasks are fixed and pre-defined.
ii) Only CPU is there. Memory, I/O chips are connected extra.	ii) CPU, Memory, I/O port are all connected on the same chip.
iii) Higher clock speed and RAM used is also higher.	iii) Lower clock speed and RAM used is also lower.
iv) Cost and power consumption is higher	iv) Cost and power consumption is comparatively lower.
v) Applications - pc, mobile etc.	v) Applications - washing machine, camera etc.

(b)

Block Diagram of a Microprocessor:



Block Diagram of a Microcontroller:



(c)

The three functionalities of a microprocessor are:

1) FETCH: Take-in binary-coded instructions from memory.

2) DECODE: Analyze or make sense of the instructions

3) EXECUTE: Carry-out the instructions. Controls overall operation of the computer.

(d)

When we say, a microprocessor has a 16-bit data-bus, it means that the microprocessor can transfer 16-bits or 2 bytes of data between the CPU and external components in a single clock-cycle.

An address bus is a unidirectional set of wires that carries memory address of the instructions which are to be executed. The information transfer takes place from the processor to Memory or I/o elements. It also determines the number of locations that CPU can address. For example, in 8086 which has 20-bit address bus, can access 2^{20} bytes or 1MB of memory.

(e)

An Intel 8086 microprocessor has a 20-bit address bus. The maximum size of the memory can be.

$$= 2^{20} \text{ bytes}$$

$$= 1 \text{ MB}$$

So, the maximum size of the memory compatible with an 8086 microprocessor is 1MB.

A 32-bit microprocessor has 32-bit address bus. The maximum size of the memory can be

$$= 2^{32} \text{ bytes}$$

$$= 4 \times 2^{30} \text{ bytes}$$

$$= 4 \text{ GB}$$

So, a 32-bit microprocessor can access 4GB of memory with its 32-bit address bus. If an 8GB RAM is used other 4GB of the RAM is unaddressable or inaccessible by a 32-bit microprocesor.

(f)

An assembler is a program that converts the assembly language into machine language. It is similar to compilers but more simplistic.

Differentiation between RAM and ROM:

i) RAM (Random Access Memory):

* Can be written on read to anytime by the CPU.
* It is a volatile memory. It means contents of RAM are erased when computer is turned off.

ii) ROM (Read-Only Memory):

* Can only be read by the CPU.
* It is pre-loaded with data and software that never changes. For example, computer's BIOS.
* It is not a volatile memory.

Question-2

Given,

CS = A38Ah
DS = 7F10h
SS = 9C45h
ES = D2B8h
SI = 4E3Ah
DI = 91F0h
SP = B27Ch
BP = 06D2h
IP = 3E2Ah
CX = 0CE7h

(a)

For physical address in Extra segment,

we know, physical address = segment no. x 10h + offset

segment no. = D2B8h

offset, DI = 91F0h

∴ physical address = (D2B8h x 10h) + 91F0h

= D2B80 + 91F0h

= DBD70h

∴ the physical address of the current index in Extra segment is DBD70h. (Ans.)

(b) (I)

No, in the 8086 architecture, a segment no. is a 16-bit address that must start on a 16-byte boundary. That means it has to be a multiple of 16h which consequently means it must end with a 0h. Since the address 0001Ah and A76CBh are 20-bit address and doesn't end with 0h, it can't be a segment number.

(b) (II)

For 0001Ah,

maximum base address = 0001 0h

∴ one logical address = 0001 : 000A

and, minimum base address = 00000h

another logical addres = 0000 : 001A

so, two valid logical address for 0001Ah are

0001 : 000A and 0000 : 001A

For A76CBh,

maximum base address = A76C0h

∴ one logical address = A76C : 000B

and, minimum base address = 976D0

∴ another logical address = 976D:FFFB

(Ans.)

(c)

In overlapping segmentation, two or more segments share a portion of the same physical memory space. On the other hand, in non-overlapping segmentation, memory segments are strictly distinct and do not share any physical memory space.

Two advantages of segmentation are:

1. It provides a powerful memory management mechanism.
2. It allows processes to easily share data.

(d)

from 'a', we have our physical address DBD70h.

for second highest logical address,

maximum base address = DBD70h

second highest base address = DBD60h

∴ 2nd highest logical address = DBD6 : 0010

for third lowest logical address,

minimum base address = CBD80h

2nd lowest base address = CBD90h

3rd lowest base address = CBDA0h

∴ 3rd lowest logical address = CBDA : FFD0 (Ans:)

(e)

Given,

MOV [IP + 0002h], CX

IP + 00002h = 3E2Ah + 0002h

= 3E2Ch

Now, physical address,

segment no., CS = A38Ah

offset = 3E2Ch

∴ physical address = (A38Ah × 10h) + 3E2Ch

= A76CCh

after the instruction, table-1 will look like

Address	0001Ah	A76CBh	A76CCh	A76CDh	A76CEh
Data	3D	A0	E7	0C	0A

Question-3

(a)

Given physical address = 111h

= 00111h

for smallest segment no. and logical address, we need the maximum possible offset. Now,

minimum base address = 00000h

∴ smallest segment no. = 0000h

smallest logical address = 0000:0111

for largest segment no. and logical address, we need the minimum possible offset. So,

maximum base address = 00110h

∴ maximum segment no. = 0011h

maximum logical address = 0011:0001

for second largest segment no. and logical address, we need the next of smallest offset. So,

maximum base address = 0011 0h

2nd largest base address = 0010 0h

∴ 2nd largest segment. no = 0010h

2nd largest logical address = 0010 : 0011

(b)

largest segment no. we found is 0011h.

So, largest base address = 0011 0h.

∴ 4th largest Physical address = 0011 0h + FFFCh

= 1010Ch

So, 4th largest physical address for the largest segment no. is 1010Ch. (Ans:)

(c)

Given, Physical address = 88292h

Since this is non-overlapping segmentation, the segment no. can be found through this,

base address = 80000h

Segment no. = 8000h

and, logical address = 8000 : 8292

So, 88292h belongs to segment 8000h and it's logical address is 8000 : 8292.

(Ans.)